# Oscar Hernandez

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## Education

**Bard College at Simon's Rock** — **Great Barrington, MA**
*B.A. Mathematics and Computer Science* — *– Spring 2017*
*A.A. Liberal Arts and Sciences* — *Fall 2013 – Spring 2015*

## Experience

**Ceres Automation** — **Riverside I/O**
*Co-founder, Automation Engineer* — *January 2018 –*

- Designing high-throughput electromechanical systems to automate additive manufacturing facilities and bioengineering research laboratories.
- Serve as the EECS (electrical engineering & computer science) lead and principal author of all software systems, guiding teams of 0-3 software engineers to write embedded software and web applications.

**VNenergy** — **ExCITE Accelerator**
*Part-time Software Engineer* — *August 2018 –*

- Implementing mathematical optimization techniques to rigorously test predictive machine learning model, and developing full-stack web application interface.

**Zap Crunch** — **Riverside I/O**
*Freelance Full-Stack Web Developer* — *February 2018–*

- Designing website for client in politics industry, and implementing Content Management System to allow non-programmers to add and edit pages.
- Developed+deployed Express.js web application for client in e-commerce industry, which syncs inventory quantities from multiple Shopify stores and Brandboom showrooms.
- Developed website for client in game industry and deployed in testing subdomain; deploying to public domain soon.

**Success Academy Charter Schools** — **Harlem, NY**
*5th Grade Associate STEM Teacher* — *July 2017 – December 2018*

- Trained to teach 5th Grade Mathematics and Earth Science at a fast-paced public middle school. Taught 5th Grade Humanities, serving as the lead History teacher and assistant ELA teacher.
- Served as the school's sole computer science teacher, designing, implementing, teaching the curricula for grades 5-8 from scratch.

**Bard College at Simon's Rock**
*Summer 2015/2016 Research Intern, Research Assistant, Senior* — *May 2015 – June 2017*

- Explored the computational complexity of the turnstile mechanic in classic games [Her+18], proving that its inclusion makes games PSPACE-hard in the general case and NP-hard in several limited cases.
- Construct simple framework [HW17] for creating all maximum-rotating $k$-ary de Bruijn sequences

of length $n$ by constructing a new one-to-one tree isomorphism.
- Published findings on dynamics of neural networks [Has+17], by using self-written simulation software to predict the behavior of analog neural networks.
- Explored equilibria of trade systems with $n$ traders, and varying distributions of goods, preferences and wealths.
- Published novel algorithms for constructing binary [DHW16] and $k$-ary [Dra+18] de Bruijn sequences.

## Computer Skills

**General**: Python, JavaScript, SQL, Shell, Haskell, Prolog, Ruby, Java, C, C++, C#;

**Design**: HTML, CSS, PHP, Markdown, LaTeX, Blender, Maya, Unity3D;

**Web**: Meteor, React, Angular, Express, Flask, Bootstrap, Jekyll, Hugo;

**Applications**: Vim, Git+GitHub, DigitalOcean, Heroku, AWS, Netlify, Octave;

**Operating Systems**: Parabola GNU/Linux-libre, Fedora, Ubuntu, Windows, macOS, Android;

**Hardware**: Raspberry Pi, Arduino, RAMBo, motors, drivers, power supply units, LEDs.

## Publications


[Dra+18] Patrick Dragon et al. "Constructing de Bruijn sequences with Co-Lexicographic Order: The $k$-ary Grandmama Sequence". In: *European Journal of Combinatorics* 72 (Aug. 2018), pp. 1–11. URL: https://www.sciencedirect.com/science/article/pii/S0195669818300696.

[HW17] Oscar Hernandez and Aaron Williams. "Constructing Maximum-Rotating de Bruijn Sequences Using Necklace Trees". thesis. Great Barrington, MA: Bard College at Simon's Rock, May 2017. URL: http://digitalcommons.bard.edu/sr-theses/1120/.

[Has+17] Harold Hastings et al. "Dynamics of Biomimetic Electronic Artificial Neural Networks". In: *Proceedings of the 4th International Conference on Applications in Nonlinear Dynamics*. Ed. by Visarath In, Patrick Longhini, and Antonio Palacios. Vol. 6. Lecture Notes in Networks and Systems. Denver, CO: Springer International Publishing AG, Mar. 2017, pp. 195–207. URL: https://link.springer.com/chapter/10.1007/978-3-319-52621-8_18.

[DHW16] Patrick Dragon, Oscar Hernandez, and Aaron Williams. "The Grandmama de Bruijn Sequence for Binary Strings". In: *LATIN 2016: Theoretical Informatics*. Ed. by Evangelos Kranakis, Gonzalo Navarro, and Edgar Chávez. Vol. 9644. Lecture Notes in Computer Science. Ensenada, Mexico: Springer-Verlag Berlin Heidelberg, Apr. 2016, pp. 347–361. URL: https://link.springer.com/chapter/10.1007/978-3-662-49529-2_26.


### Manuscripts


[Her+18] O. Hernandez et al. "Spinning Around and Around: Turnstile Puzzles are PSPACE-Complete". Apr. 2018.


## Interests

**Current**: Robotic Process Automation, Artificial Intelligence, Parabola GNU/Linux-libre, Judaism

**Prospective**: Prolog, Google Apps Script, Browser Extensions, Native Mobile Apps, Perl